

Program: Diploma in Renewable Energy	
Course Code : 5292	Course Title: Fundamentals of Power Electronics
Semester : 3	Credits : 4
Course Category : Program Core	
Periods per week: 4 (L: 3 T: 1 P: 0)	Periods per semester : 60

Course Objectives:

- To impart basic knowledge of power electronics.
- To understand the fundamental concepts of different power electronics converters.
- To familiarize the applications of power electronic devices.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic concepts of Electrical & Electronics Engineering		Fundamentals of Electrical & Electronics Engineering	2
Basic concepts and Operations of Electronic Devices		Analog & Opto Electronics	3

Course Outcomes:

On completion of the course, the students will be able to:-

COn.	Description	Duration (Hours)	Cognitive Level
CO 1	Identify the operation of different power electronic devices	15	Applying
CO 2	Identify the operation of single phase rectifier circuits.	16	Applying

CO 3	Identify the working of chopper, inverter and AC voltage converters.	15	Applying
CO 4	Summarize the concepts of electric drives.	12	Understanding
	Series Test	2	

CO – PO Mapping

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	2						
CO2	3						
CO3	3						
CO4	2						

3- Strongly mapped

2- Moderately mapped

1- Weakly mapped

Course Outline

On completion of the course, the students will be able to:-

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Identify the operation of different power electronic devices		
M1.01	Summarize the fundamental concepts of power electronic circuits.	2	Understanding
M1.02	Explain the Operation and applications of various power electronic devices.	6	Understanding
M1.03	Illustrate the working and characteristics of SCR.	6	Understanding
M1.04	List the applications and specifications of SCR	1	Understanding

Contents:

Concept of Power electronics: Definition -Block Diagram-Scope and Applications -Types of Power Electronic Circuits.

Power electronic devices: Symbols - Structure- Operation -Characteristics-types - applications- Power diode- Power Transistor-DIAC-TRIAC- GTO- MOSFET- IGBT

Silicon Controlled Rectifier (SCR): Symbol - Structure- Operation - V-I Characteristics- Two transistor analogy of SCR- Dynamic characteristics-Turn on and Turn off characteristics (explanation not needed)-

Applications and Specifications of SCR - List various applications and specifications of SCR

CO2	Identify the operation of single phase rectifier circuits.		
M2.01	Outline various methods of turn-on and turn-off of SCR	3	Understanding
M2.02	Explain different protection methods of SCR.	2	Understanding
M2.03	Identify the waveforms of single phase half wave controlled rectifier with various loads	6	Applying
M2.04	Identify the waveforms of single phase full wave controlled rectifiers with various loads.	5	Applying
	Series Test – I	1	
Turn on and Turn off Methods of SCR -Turn-On methods - basic concepts - gate triggering (R, RC, UJT triggering circuits) -forward voltage triggering- resistance triggering temperature triggering-light triggering- dv/dt triggering-SCR Turn-Off methods-basic concepts -Line commutation -Forced commutation - Resonant pulse commutation- Load Commutation Thyristor protection circuits -Over voltage protection-dv/dt protection,Over current protection-di/dt protection, Gate protection-Snubber Circuit Phase Controlled Rectifiers: Basic principle of phase control operation of converters- Phase control parameters-Firing angle & conduction angle-circuit ,operation and waveforms of single phase half wave controlled rectifier-with R & RL loads (with and without Freewheeling Diode) Single phase full wave controlled rectifiers -Circuit ,operation and waveforms of single phase full wave controlled centre tapped and bridge rectifiers with R & RL loads.			
CO3	Identify the working of chopper, inverter and AC voltage converters.		

M3.01	Illustrate the operation of chopper circuits	3	Applying
M3.02	Explain the operation of different single phase inverters	5	Understanding
M3.03	Summarize the operation of cyclo converters .	4	Understanding
M3.04	Outline the working of AC Voltage Controllers with R and RL loads.	2	Understanding
Contents : DC Choppers : Principle of step down and step up chopper operation-Control strategy of chopper- Chopper configuration and quadrant of operation- Type A, B, C, D and E chopper INVERTER: Introduction- Inverter classification-Circuit ,operation and waveforms of Voltage source series inverter,Single phase Voltage source half and full Bridge Inverter with resistive load – Pulse width modulation (basic concept only) CYCLO CONVERTER: Principle of Cyclo-converter operation- Single phase to single phase operations - step up and step down (Circuit ,operation and waveforms only) AC voltage controllers (ACVC) : Single phase full-wave ACVC with R, & RL loads (working , diagram and waveforms)			
CO4	Summarize the concepts of electric drives.		
M4.01	Explain the working of Buck, Boost and Buck- Boost converters.	3	Understanding
M4.02	Outline the concepts of electric drives.	2	Understanding
M4.03	Summarize different speed control of dc motor drives	4	Understanding
M4.04	Explain different speed control of Induction motor drives	4	Understanding
	Series Test – II	1	

Contents :

Switching regulators – Buck, Boost & Buck-boost –Circuit diagram & Operation.

Electric Drive: Introduction to electric drives – Block diagram – advantages of electric drives- types of load – classification of load torque

Speed control of dc motor drives : methods of speed control of D.C. Motor - circuit diagram and working -armature voltage control- chopper control

Speed control of induction motor drives: methods of speed control of induction motor -circuit diagram and working- stator voltage control, frequency control, V/F control.

Text / Reference

T/R	Book Title/Author
T1	P.S. Bimbhra, Power Electronics, Khanna Publishers, New Delhi
T2	Singh M. D. and K. B. Khanchandani, Power Electronics, Tata McGraw Hill, New Delhi, 2008.
T3	Muhammad H. Rashid, Power Electronics Circuits, Devices and Applications, Pearson Education
T4	Daniel W. Hart, Power Electronics, Tata McGraw-Hill Education
R1	Mohan N., T. M. Undeland and W. P. Robbins., Power Electronics, Converters, Applications & Design, Wiley-India
R2	Fundamentals of Power Electronics, Erickson, Robert W., and Maksimovic, Dragan.
R3	Krein P. T., Elements of Power Electronics, Oxford University Press, 1998.

Online Resources

Sl.No	Website Link
1	https://btechgeeks.com/power-electronics-notes/
2	https://www.electrical4u.com/concept-of-power-electronics/
3	https://www.electronicshub.org/scr-turn-on-methods/
4	electricalworkbook.com/controlled-rectifier/
5	https://www.elprocus.com/a-brief-introduction-to-chopper-circuits/
6	ircuitdigest.com/tutorial/cycloconverter-types-working-circuits-applications
7	https://electricalworkbook.com/ac-voltage-controller/
8	https://www.maximintegrated.com/en/products/power/switching-regulators.html
9	https://www.electrical4u.com/electrical-drives/
10	https://www.tutorialspoint.com/difference-between-ac-drives-and-dc-drives